

Supplementary Material — Group-conformal calibration of near-infrared prediction intervals for new seasons, instruments and regions using PLS diagnostics

[Authors]

Abstract

Supplementary tables S1–S15 and figures S1–S2. All values are generated from the result files of the accompanying code (run_all.py).

Keywords: Supplementary material

S1. Method settings

Table S1. Fixed settings of all experiments (config/experiment_config.json).

Component	Setting
seeds	[0, 1, 2, 3, 4]
alphas	[0.05, 0.1, 0.2]
primary_alpha	0.1
preprocessing	{mango: Savitzky-Golay 2nd derivative, window 13, poly 2, wavelengths 684-990 nm, ossl: log10(1/R) at 10 nm steps 400-2500 nm, Savitzky-Golay 1st derivative, window 11; target log(1+OC), corn: SNV then Savitzky-Golay 1st derivative, window 15, tablets: SNV then Savitzky-Golay 1st derivative, window 15}
pls	{A_max: {mango: 20, ossl: 25, corn: 15, tablets: 15}, selection: min group-balanced RMSE of out-of-group cross-validation}
split_conformal	{calibration_fraction: 0.25, split_unit: physical unit}
random_cv_folds	5
scale_model	{features: [log(1+T2), log(Q/Q_ref)], link: log, loss: Poisson quasi-likelihood, floor_frac: 0.1}
gpr	{subset: 1000, kernel: Constant*RBF(ARD)+White}
bagging	{B: 25, bootstrap_unit: physical unit}
quantile_gbm	{n_estimators: 300, learning_rate: 0.05, num_leaves: 31}
wcp	{density_ratio: logistic regression on standardised PLS scores and log Q, max_n_per_class: 20000}
lucas_spatial_blocks	{k: 30, method: k-means on (lat, lon*cos(lat)), seed: 0}
cnv	{architecture: Conv1d(1->8, kernel 9, same padding) - ELU - Flatten - Linear(64) - ELU - Linear(32) - ELU - Linear(1), inputs: per-wavelength standardised (training statistics), optimizer: Adam lr 1e-3, weight decay 1e-4, batch 256, epochs: max 40, early stopping patience 8 on a 10 % unit-grouped internal validation split, device: auto (CPU/CUDA/MPS)}
wcp_clip	{weights: density ratios normalised to mean 1 on the calibration set and capped at 20}

S2. Generalisation scenarios and exchangeable control

Table S2. Coverage (cov.) and interval score (IS) at nominal 0.90 for the LUCAS campaign shift (2009 ↔ 2015), the KSSL → LUCAS library transfer, and the tablet data (calibration on instrument 1; test on instrument 1 = exchangeable control, instrument 2 = instrument shift without group information). Means over five seeds.

Method	S-campaign cov.	S-campaign IS	S- KSSL → LUC AS cov.	S- KSSL → LUC AS IS	Tablets inst. 1 cov.	Tablets inst. 2 cov.
ASTM-R	0.861	13.137	0.053	136.577	0.974	0.890
ASTM-G	0.876	12.788	0.071	126.234	0.974	0.890
BAG	0.861	13.154	0.055	136.924	0.976	0.910
GPR	0.802	12.546	0.975	83.180	0.962	0.749
CQR	0.874	9.760	0.310	105.562	0.920	0.961
SCP	0.847	13.591	0.046	148.382	0.932	0.760
CV+	0.852	13.352	0.050	139.221	0.948	0.831
WCP	0.803	∞	1.000	∞	0.923	1.000
G-W-A	0.869	12.945	0.066	128.916	0.940	0.795
GC-D	0.892	12.248	0.604	95.520	0.936	0.889
GC-D+	0.924	11.856	0.663	99.205	0.940	0.893
HJ+	0.908	12.169	0.078	122.659	0.948	0.831

Method	S-campaign cov.	S-campaign IS	S- KSSL → LUC AS cov.	S- KSSL → LUC AS IS	Tablets inst. 1 cov.	Tablets inst. 2 cov.
HCP-D	0.955	12.715	0.712	101.500	0.947	0.926
ORACLE	0.902	11.382	0.902	134.476	0.901	0.902

S3. Other nominal levels

Table S3a. Group-averaged coverage at nominal 0.95.

Method	M- season	M- forward	M- instrum ent	M- populati on	S-region	C-moist	C-oil	C-prot	C- starch	S- campaig n	S- KSSL → LUCAS	Tablets
ASTM-R	0.893	0.860	0.871	0.939	0.932	0.649	0.883	0.773	0.661	0.911	0.089	0.950
ASTM-G	0.945	0.929	0.933	0.946	0.943	0.653	0.933	0.812	0.707	0.922	0.127	0.950
BAG	0.893	0.862	0.874	0.939	0.933	0.658	0.926	0.814	0.736	0.911	0.094	0.963
GPR	0.866	0.828	0.828	0.925	0.928	0.719	0.883	0.819	0.697	0.863	0.994	0.909
QGB	0.827	0.764	0.819	0.867	0.901	0.812	0.782	0.767	0.757	0.884	0.315	0.878
CQR	0.875	0.827	0.871	0.909	0.942	0.958	0.966	0.966	0.948	0.932	0.337	0.971
SCP	0.896	0.865	0.873	0.941	0.940	0.663	0.888	0.732	0.708	0.915	0.092	0.905
NCP-kNN	0.857	0.813	0.819	0.909	0.932	0.656	0.888	0.739	0.701	0.879	0.201	0.957
CV+	0.896	0.862	0.874	0.941	0.940	0.654	0.890	0.760	0.698	0.917	0.100	0.942
WCP	0.930	0.925	0.939	0.950	0.942	0.986	0.973	0.997	0.970	0.833	1.000	0.980
HJ+	1.000	1.000	0.977	0.955	0.982	1.000	1.000	1.000	1.000	0.972	0.214	0.942
HCP-D	1.000	1.000	1.000	0.961	1.000	1.000	1.000	1.000	1.000	1.000	0.948	0.984
GC-D+	1.000	1.000	0.972	0.954	0.982	1.000	1.000	1.000	1.000	0.978	0.926	0.963
GC-D	0.947	0.933	0.934	0.949	0.949	0.569	0.834	0.777	0.742	0.943	0.878	0.959
ORACLE	0.949	0.949	0.950	0.948	0.952	1.000	1.000	1.000	1.000	0.953	0.952	0.952

Table S3b. Group-averaged coverage at nominal 0.80.

Method	M- season	M- forward	M- instrum ent	M- populati on	S-region	C-moist	C-oil	C-prot	C- starch	S- campaig n	S- KSSL → LUCAS	Tablets
ASTM-R	0.732	0.674	0.688	0.803	0.807	0.583	0.727	0.556	0.518	0.767	0.028	0.877
ASTM-G	0.821	0.790	0.789	0.819	0.827	0.587	0.783	0.621	0.568	0.788	0.036	0.877
BAG	0.731	0.678	0.690	0.803	0.809	0.583	0.761	0.586	0.547	0.765	0.029	0.896
GPR	0.685	0.634	0.640	0.779	0.800	0.610	0.709	0.600	0.548	0.699	0.916	0.768
QGB	0.607	0.526	0.572	0.676	0.721	0.610	0.625	0.610	0.565	0.691	0.205	0.734
CQR	0.659	0.587	0.627	0.730	0.779	0.784	0.781	0.824	0.736	0.767	0.222	0.877
SCP	0.714	0.661	0.667	0.785	0.778	0.558	0.724	0.576	0.522	0.725	0.021	0.711
NCP-kNN	0.686	0.640	0.645	0.750	0.771	0.553	0.725	0.579	0.531	0.679	0.062	0.802
CV+	0.713	0.658	0.669	0.785	0.779	0.573	0.739	0.564	0.526	0.735	0.024	0.785
WCP	0.785	0.786	0.839	0.825	0.866	0.927	0.896	0.967	0.870	0.761	1.000	0.922
HJ+	0.934	0.978	0.819	0.806	0.827	1.000	1.000	1.000	1.000	0.789	0.028	0.785
HCP-D	1.000	1.000	0.829	0.811	0.854	1.000	1.000	1.000	1.000	0.837	0.357	0.857
GC-D+	0.929	0.968	0.804	0.804	0.824	1.000	1.000	1.000	1.000	0.816	0.242	0.834
GC-D	0.797	0.770	0.776	0.799	0.798	0.458	0.641	0.621	0.602	0.787	0.219	0.821

Method	M-season	M-forward	M-instrument	M-population	S-region	C-moist	C-oil	C-prot	C-starch	S-campaign	S-KSSL → LUCAS	Tablets
ORACLE	0.799	0.798	0.800	0.800	0.802	0.826	0.815	0.812	0.813	0.802	0.801	0.801

S4. Worst-group coverage and share of under-covered groups

Table S4a. Lowest coverage among held-out groups (nominal 0.90).

Method	M-season	M-forward	M-instrument	M-population	S-region	C-moist	C-oil	C-prot	C-starch	S-campaign	S-KSSL → LUCAS	Tablets
ASTM-R	0.733	0.699	0.251	0.013	0.764	0.000	0.270	0.090	0.000	0.644	0.006	0.890
ASTM-G	0.798	0.811	0.336	0.013	0.794	0.000	0.530	0.120	0.000	0.659	0.010	0.890
BAG	0.733	0.703	0.258	0.013	0.768	0.000	0.540	0.150	0.000	0.645	0.006	0.910
GPR	0.659	0.653	0.157	0.010	0.758	0.090	0.550	0.150	0.000	0.442	0.879	0.749
QGB	0.616	0.565	0.303	0.013	0.752	0.560	0.560	0.480	0.340	0.647	0.067	0.759
CQR	0.675	0.644	0.346	0.037	0.816	0.820	0.700	0.770	0.490	0.739	0.075	0.920
SCP	0.721	0.695	0.236	0.013	0.759	0.000	0.220	0.060	0.000	0.627	0.004	0.760
NCP-kNN	0.674	0.581	0.137	0.345	0.791	0.000	0.230	0.090	0.000	0.585	0.020	0.865
CV+	0.724	0.692	0.244	0.013	0.757	0.000	0.400	0.080	0.000	0.632	0.006	0.831
WCP	0.717	0.682	0.567	0.028	0.599	0.830	0.800	0.950	0.700	0.560	1.000	0.923
HJ+	1.000	1.000	0.452	0.013	0.843	1.000	1.000	1.000	1.000	0.720	0.012	0.831
HCP-D	1.000	1.000	0.600	0.015	0.885	1.000	1.000	1.000	1.000	0.809	0.272	0.926
GC-D+	1.000	1.000	0.447	0.013	0.837	1.000	1.000	1.000	1.000	0.720	0.114	0.893
GC-D	0.798	0.772	0.356	0.013	0.786	0.000	0.330	0.140	0.020	0.649	0.092	0.889
ORACLE	0.896	0.896	0.897	0.868	0.897	0.880	0.840	0.840	0.870	0.896	0.898	0.901

Table S4b. Share of held-out groups with coverage below 0.85 (nominal 0.90).

Method	M-season	M-forward	M-instrument	M-population	S-region	C-moist	C-oil	C-prot	C-starch	S-campaign	S-KSSL → LUCAS	Tablets
ASTM-R	0.57	0.80	0.46	0.23	0.27	0.37	0.40	0.53	0.43	0.40	1.00	0.00
ASTM-G	0.14	0.40	0.17	0.18	0.23	0.33	0.23	0.43	0.33	0.34	1.00	0.00
BAG	0.57	0.80	0.46	0.22	0.27	0.33	0.30	0.50	0.33	0.40	1.00	0.00
GPR	0.71	0.80	0.62	0.26	0.30	0.37	0.43	0.63	0.70	0.59	0.00	0.50
QGB	1.00	1.00	0.92	0.55	0.60	0.90	0.90	0.90	0.97	0.78	1.00	0.50
CQR	0.57	1.00	0.67	0.35	0.17	0.07	0.20	0.13	0.47	0.24	1.00	0.00
SCP	0.71	0.80	0.54	0.24	0.27	0.37	0.60	0.57	0.63	0.45	1.00	0.50
NCP-kNN	0.71	1.00	0.75	0.37	0.27	0.40	0.47	0.50	0.60	0.76	1.00	0.00
CV+	0.71	0.80	0.58	0.24	0.27	0.37	0.43	0.53	0.47	0.41	1.00	0.50
WCP	0.29	0.20	0.21	0.16	0.10	0.03	0.07	0.00	0.13	0.64	0.00	0.00
HJ+	0.00	0.00	0.08	0.18	0.07	0.00	0.00	0.00	0.00	0.19	1.00	0.50
HCP-D	0.00	0.00	0.08	0.17	0.00	0.00	0.00	0.00	0.00	0.02	0.90	0.00
GC-D+	0.00	0.00	0.17	0.18	0.13	0.00	0.00	0.00	0.00	0.09	0.83	0.00
GC-D	0.29	0.40	0.29	0.19	0.27	0.63	0.57	0.67	0.53	0.29	0.90	0.00
ORACLE	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.03	0.00	0.00	0.00	0.00

S5. Sensitivity analysis

Table S5. Sensitivity of coverage, width and interval score to the number of latent variables (A_{offset}), the features of the scale model (diag_mode), the floor of σ (floor_frac), the preprocessing, and the number of population folds (nominal 0.90, seed 0).

task	variant	method	coverage	width	interval_score	A
mango_population	group_folds=20	G-W-A	0.900	3.694	4.959	19.201
mango_population	group_folds=20	GC-D	0.898	3.680	4.921	19.201
mango_population	group_folds=20	R-U-D	0.891	3.571	4.920	19.201
mango_population	group_folds=20	SCP	0.885	3.530	4.977	19.201
mango_population	group_folds=5	G-W-A	0.901	3.705	4.955	19.201
mango_population	group_folds=5	GC-D	0.900	3.692	4.920	19.201
mango_population	group_folds=5	R-U-D	0.891	3.569	4.906	19.201
mango_population	group_folds=5	SCP	0.885	3.532	4.969	19.201
mango_season_loso	$A_{\text{offset}}=-4$	G-W-A	0.900	4.832	6.297	10.857
mango_season_loso	$A_{\text{offset}}=-4$	GC-D	0.894	4.836	6.249	10.857
mango_season_loso	$A_{\text{offset}}=-4$	R-U-D	0.852	4.203	6.309	10.857
mango_season_loso	$A_{\text{offset}}=-4$	SCP	0.837	4.051	6.484	10.857
mango_season_loso	$A_{\text{offset}}=4$	G-W-A	0.894	4.532	6.313	18.286
mango_season_loso	$A_{\text{offset}}=4$	GC-D	0.884	4.498	6.081	18.286
mango_season_loso	$A_{\text{offset}}=4$	R-U-D	0.825	3.717	6.406	18.286
mango_season_loso	$A_{\text{offset}}=4$	SCP	0.806	3.515	6.719	18.286
mango_season_loso	diag_mode=Q	G-W-A	0.899	4.507	6.157	14.857
mango_season_loso	diag_mode=Q	GC-D	0.898	4.561	6.003	14.857
mango_season_loso	diag_mode=Q	R-U-D	0.840	3.789	6.194	14.857
mango_season_loso	diag_mode=Q	SCP	0.823	3.602	6.395	14.857
mango_season_loso	diag_mode=T 2	G-W-A	0.899	4.507	6.157	14.857
mango_season_loso	diag_mode=T 2	GC-D	0.894	4.512	5.939	14.857
mango_season_loso	diag_mode=T 2	R-U-D	0.829	3.706	6.241	14.857
mango_season_loso	diag_mode=T 2	SCP	0.823	3.602	6.395	14.857
mango_season_loso	floor_frac=0.0 2	G-W-A	0.899	4.507	6.157	14.857
mango_season_loso	floor_frac=0.0 2	GC-D	0.895	4.519	5.900	14.857
mango_season_loso	floor_frac=0.0	R-U-D	0.839	3.787	6.129	14.857

task	variant	method	coverage	width	interval_score	A
	2					
mango_season_loso	floor_frac=0.0	SCP	0.823	3.602	6.395	14.857
	2					
mango_season_loso	floor_frac=0.3	G-W-A	0.899	4.507	6.157	14.857
mango_season_loso	floor_frac=0.3	GC-D	0.895	4.519	5.900	14.857
mango_season_loso	floor_frac=0.3	R-U-D	0.839	3.787	6.129	14.857
mango_season_loso	floor_frac=0.3	SCP	0.823	3.602	6.395	14.857
mango_season_loso	preprocessing=SNV+SG1	G-W-A	0.898	4.893	6.549	19.286
mango_season_loso	preprocessing=SNV+SG1	GC-D	0.893	4.899	6.501	19.286
mango_season_loso	preprocessing=SNV+SG1	R-U-D	0.833	4.073	6.710	19.286
mango_season_loso	preprocessing=SNV+SG1	SCP	0.823	3.961	6.838	19.286
ossl_lucas_block	A_offset=-4	G-W-A	0.900	6.545	12.883	20.900
ossl_lucas_block	A_offset=-4	GC-D	0.898	7.272	12.246	20.900
ossl_lucas_block	A_offset=-4	R-U-D	0.891	7.076	12.307	20.900
ossl_lucas_block	A_offset=-4	SCP	0.886	6.163	13.088	20.900
ossl_lucas_block	A_offset=4	G-W-A	0.899	6.467	12.483	25.000
ossl_lucas_block	A_offset=4	GC-D	0.899	7.140	11.893	25.000
ossl_lucas_block	A_offset=4	R-U-D	0.889	6.849	11.988	25.000
ossl_lucas_block	A_offset=4	SCP	0.883	6.083	12.850	25.000
ossl_lucas_block	diag_mode=Q	G-W-A	0.899	6.470	12.496	24.900
ossl_lucas_block	diag_mode=Q	GC-D	0.898	6.951	11.979	24.900
ossl_lucas_block	diag_mode=Q	R-U-D	0.888	6.646	12.123	24.900
ossl_lucas_block	diag_mode=Q	SCP	0.883	6.086	12.855	24.900
ossl_lucas_block	diag_mode=T	G-W-A	0.899	6.470	12.496	24.900
	2					
ossl_lucas_block	diag_mode=T	GC-D	0.899	7.139	11.902	24.900
	2					
ossl_lucas_block	diag_mode=T	R-U-D	0.889	6.845	12.009	24.900
	2					
ossl_lucas_block	diag_mode=T	SCP	0.883	6.086	12.855	24.900
	2					
ossl_lucas_block	floor_frac=0.0	G-W-A	0.899	6.470	12.496	24.900
	2					
ossl_lucas_block	floor_frac=0.0	GC-D	0.898	7.142	11.900	24.900
	2					
ossl_lucas_block	floor_frac=0.0	R-U-D	0.889	6.855	11.998	24.900
	2					
ossl_lucas_block	floor_frac=0.0	SCP	0.883	6.086	12.855	24.900
	2					
ossl_lucas_block	floor_frac=0.3	G-W-A	0.899	6.470	12.496	24.900
ossl_lucas_block	floor_frac=0.3	GC-D	0.898	7.142	11.900	24.900
ossl_lucas_block	floor_frac=0.3	R-U-D	0.889	6.855	11.998	24.900
ossl_lucas_block	floor_frac=0.3	SCP	0.883	6.086	12.855	24.900

S6. Convolutional network as point predictor

Table S6. 1D-CNN point predictor (mango, new season; group-averaged coverage over seven seasons; mean over seeds). $\sigma(x)$ uses the PLS diagnostics of the same spectra.

method	coverage	coverage_mi n	width	interval_sco re	rmse	n_seasons	n_seeds
CNN-G-W-A	0.881	0.664	4.606	6.293	1.403	7	3
CNN-GC-D	0.886	0.706	4.563	6.106	1.403	7	3
CNN-R-U-A	0.646	0.401	2.523	8.483	1.403	7	3
CNN-SCP	0.651	0.442	2.533	8.297	1.393	7	3

S7. Post-hoc iteration (GC-D2, GC-CQR)

Table S7a. Post-hoc variants (seeds 0–2; nominal 0.90).

task	method	coverage	coverage_mi n	width	interval_scor e	n_groups
corn_moisture	CQR	0.901	0.817	1.191	1.528	30
corn_moisture	GC-CQR	0.814	0.650	0.959	1.547	30
corn_moisture	GC-D	0.529	0.000	0.879	4.753	30
corn_moisture	GC-D2	0.538	0.000	0.851	4.951	30
corn_moisture	SCP	0.612	0.000	0.994	4.635	30
corn_oil	CQR	0.889	0.683	0.522	0.642	30
corn_oil	GC-CQR	0.877	0.667	0.463	0.581	30
corn_oil	GC-D	0.766	0.317	0.406	0.760	30
corn_oil	GC-D2	0.752	0.317	0.396	0.774	30
corn_oil	SCP	0.810	0.350	0.455	0.855	30
corn_protein	CQR	0.927	0.767	1.310	1.514	30
corn_protein	GC-CQR	0.898	0.633	1.137	1.414	30
corn_protein	GC-D	0.710	0.083	1.098	2.119	30
corn_protein	GC-D2	0.673	0.117	0.967	2.257	30
corn_protein	SCP	0.664	0.000	0.970	2.606	30
corn_starch	CQR	0.836	0.417	2.263	3.395	30
corn_starch	GC-CQR	0.827	0.500	2.031	3.283	30
corn_starch	GC-D	0.691	0.000	2.757	6.116	30
corn_starch	GC-D2	0.677	0.000	2.559	6.017	30
corn_starch	SCP	0.600	0.000	1.983	7.345	30
mango_instrument	CQR	0.770	0.363	3.621	7.030	24
mango_instrument	GC-CQR	0.900	0.459	4.946	6.360	24
mango_instrument	GC-D	0.879	0.356	4.542	6.211	24
mango_instrument	GC-D2	0.879	0.312	4.568	6.302	24
mango_instrument	SCP	0.793	0.239	3.647	6.951	24
mango_population	CQR	0.840	0.042	3.362	5.236	199
mango_population	GC-CQR	0.903	0.108	3.879	4.980	199
mango_population	GC-D	0.899	0.013	3.686	4.932	199
mango_population	GC-D2	0.898	0.013	3.665	4.915	199

task	method	coverage	coverage_mi n	width	interval_scor e	n_groups
mango_population	SCP	0.886	0.013	3.538	4.971	199
mango_season_forward	CQR	0.727	0.642	3.470	7.474	5
mango_season_forward	GC-CQR	0.858	0.790	4.560	6.366	5
mango_season_forward	GC-D	0.877	0.772	4.526	6.143	5
mango_season_forward	GC-D2	0.879	0.765	4.532	6.136	5
mango_season_forward	SCP	0.780	0.687	3.458	7.241	5
mango_season_loso	CQR	0.788	0.673	3.532	6.405	7
mango_season_loso	GC-CQR	0.901	0.803	4.665	5.877	7
mango_season_loso	GC-D	0.895	0.798	4.519	5.900	7
mango_season_loso	GC-D2	0.892	0.765	4.509	5.974	7
mango_season_loso	SCP	0.825	0.722	3.632	6.447	7
ossl_lucas_block	CQR	0.886	0.814	5.520	8.038	30
ossl_lucas_block	GC-CQR	0.902	0.834	5.612	7.767	30
ossl_lucas_block	GC-D	0.898	0.786	7.142	11.900	30
ossl_lucas_block	GC-D2	0.898	0.789	10.430	13.529	30
ossl_lucas_block	SCP	0.884	0.760	6.110	12.777	30

Table S7b. Paired Wilcoxon tests over held-out groups (Holm across scenarios for each contrast).

task	method	comparator	n_groups	IS_median_ diff	IS_r_rb	IS_p	IS_p_holm
corn_moisture	GC-D2	GC-D	30	-0.0078	0.0151	0.9426	1.0000
corn_moisture	GC-CQR	CQR	30	-0.0408	0.0194	0.9263	0.9263
corn_moisture	GC-CQR	GC-D	30	-0.5161	-0.7634	0.0003	0.0016
corn_oil	GC-D2	GC-D	30	-0.0025	-0.1570	0.4528	1.0000
corn_oil	GC-CQR	CQR	30	-0.0506	-0.6946	0.0009	0.0063
corn_oil	GC-CQR	GC-D	30	-0.1157	-0.7763	0.0002	0.0014
corn_protein	GC-D2	GC-D	30	0.0294	0.4323	0.0387	0.3098
corn_protein	GC-CQR	CQR	30	-0.0987	-0.7333	0.0005	0.0036
corn_protein	GC-CQR	GC-D	30	-0.3574	-0.8237	0.0001	0.0007
corn_starch	GC-D2	GC-D	30	-0.1305	-0.4323	0.0387	0.3098
corn_starch	GC-CQR	CQR	30	-0.1988	-0.2774	0.1846	0.6250
corn_starch	GC-CQR	GC-D	30	-0.8402	-0.7247	0.0005	0.0026
mango_instrument	GC-D2	GC-D	24	0.0035	0.3400	0.1515	0.9090
mango_instrument	GC-CQR	CQR	24	-0.3300	-0.6133	0.0072	0.0431
mango_instrument	GC-CQR	GC-D	24	0.3357	0.4800	0.0395	0.1579
mango_population	GC-D2	GC-D	199	-0.0314	-0.2258	0.0057	0.0516
mango_population	GC-CQR	CQR	199	0.0439	-0.1082	0.1855	0.6250
mango_population	GC-CQR	GC-D	199	0.0676	0.1305	0.1106	0.3317
mango_season_forward	GC-D2	GC-D	5	-0.0145	-0.0667	1.0000	1.0000
mango_season_forward	GC-CQR	CQR	5	-0.9971	-1.0000	0.0625	0.3125
mango_season_forward	GC-CQR	GC-D	5	0.1743	0.7333	0.1875	0.3750
mango_season_loso	GC-D2	GC-D	7	0.0036	0.2143	0.6875	1.0000
mango_season_loso	GC-CQR	CQR	7	-0.7655	-0.6429	0.1562	0.6250
mango_season_loso	GC-CQR	GC-D	7	-0.0328	-0.0714	0.9375	0.9375

task	method	comparator	n_groups	IS_median_diff	IS_r_rb	IS_p	IS_p_holm
ossl_lucas_block	GC-D2	GC-D	30	-0.1455	-0.1484	0.4779	1.0000
ossl_lucas_block	GC-CQR	CQR	30	-0.1405	-0.9828	0.0000	0.0000
ossl_lucas_block	GC-CQR	GC-D	30	-1.6531	-1.0000	0.0000	0.0000

S8. Weighted conformal prediction with clipped density ratios

Table S8. WCP vs WCP-clip (density ratios normalised to mean 1 on the calibration set and capped at 20); finite_frac = share of test spectra with a bounded interval.

task	method	coverage	width_median	interval_score	finite_frac	n_groups	n_seeds
			an	re			
corn_moisture	WCP	0.964	∞	∞	0.535	30	5
corn_moisture	WCP-clip	0.964	∞	∞	0.535	30	5
corn_oil	WCP	0.945	∞	∞	0.502	30	5
corn_oil	WCP-clip	0.945	∞	∞	0.502	30	5
corn_protein	WCP	0.989	∞	∞	0.243	30	5
corn_protein	WCP-clip	0.989	∞	∞	0.243	30	5
corn_starch	WCP	0.936	∞	∞	0.539	30	5
corn_starch	WCP-clip	0.936	∞	∞	0.539	30	5
mango_instrument	WCP	0.905	∞	∞	0.805	24	5
mango_instrument	WCP-clip	0.851	4.106	∞	0.990	24	5
mango_population	WCP	0.907	∞	∞	0.895	199	5
mango_population	WCP-clip	0.899	3.595	∞	1.000	199	5
mango_season_forward	WCP	0.875	10.294	∞	0.999	5	5
mango_season_forward	WCP-clip	0.822	3.664	7.306	1.000	5	5
mango_season_loso	WCP	0.880	3.886	∞	1.000	7	5
mango_season_loso	WCP-clip	0.855	3.946	6.258	1.000	7	5
ossl_kssl_to_lucas	WCP	1.000	∞	∞	0.000	30	5
ossl_kssl_to_lucas	WCP-clip	0.106	31.198	119.848	1.000	30	5
ossl_lucas_block	WCP	0.917	∞	∞	0.980	30	5
ossl_lucas_block	WCP-clip	0.921	4.830	∞	1.000	30	5
ossl_lucas_campaign	WCP	0.803	∞	∞	0.790	58	5
ossl_lucas_campaign	WCP-clip	0.905	∞	∞	0.887	58	5
tablets	WCP	0.962	∞	∞	0.500	2	5
tablets	WCP-clip	0.962	∞	∞	0.500	2	5

S9. Computing cost

Table S9. Isolated computing cost per held-out season (mango LOSO, ~70 000 training spectra, one CPU core, mean of three seasons).

method	method_s	method_s_sd	peak_MiB	ms_per_test_spectrum	shared_setup_s
ASTM-G	0.002	0.000	2.242	0.000	4.895
G-U-A	0.004	0.000	4.575	0.000	4.742
G-W-A	0.012	0.001	5.675	0.001	4.695

method	method_s	method_s_sd	peak_MiB	ms_per_test_spectrum	shared_setup_s
ORACLE	0.020	0.011	0.747	0.002	5.190
GC-D+	0.117	0.011	83.338	0.014	4.670
G-U-D	0.120	0.008	83.338	0.015	4.887
GC-D	0.120	0.005	83.338	0.014	4.746
GC-D2	0.146	0.009	82.218	0.017	5.120
HJ+	0.566	0.022	33.700	0.066	4.766
HJ+-D	0.637	0.021	84.985	0.074	4.886
WCP	0.688	0.020	171.217	0.082	5.155
SCP	0.692	0.069	171.217	0.084	5.315
WCP-clip	0.713	0.032	171.217	0.082	5.093
HCP-A	0.771	0.067	132.092	0.092	5.210
HCP-D	1.213	0.132	136.379	0.140	5.085
R-U-A	2.936	0.223	201.935	0.358	4.914
ASTM-R	3.161	0.378	201.935	0.393	5.499
R-U-D	3.236	0.370	201.935	0.400	4.846
CV+	3.603	0.034	201.935	0.421	5.281
CQR	5.213	0.239	171.217	0.613	5.218
QGB	5.257	0.361	171.573	0.615	4.832
NCP-kNN	8.933	0.579	179.120	1.028	5.015
BAG	14.980	2.041	198.978	1.791	4.685
GPR	15.613	4.368	448.503	1.700	5.002
GC-CQR	34.502	2.496	216.729	4.156	4.882

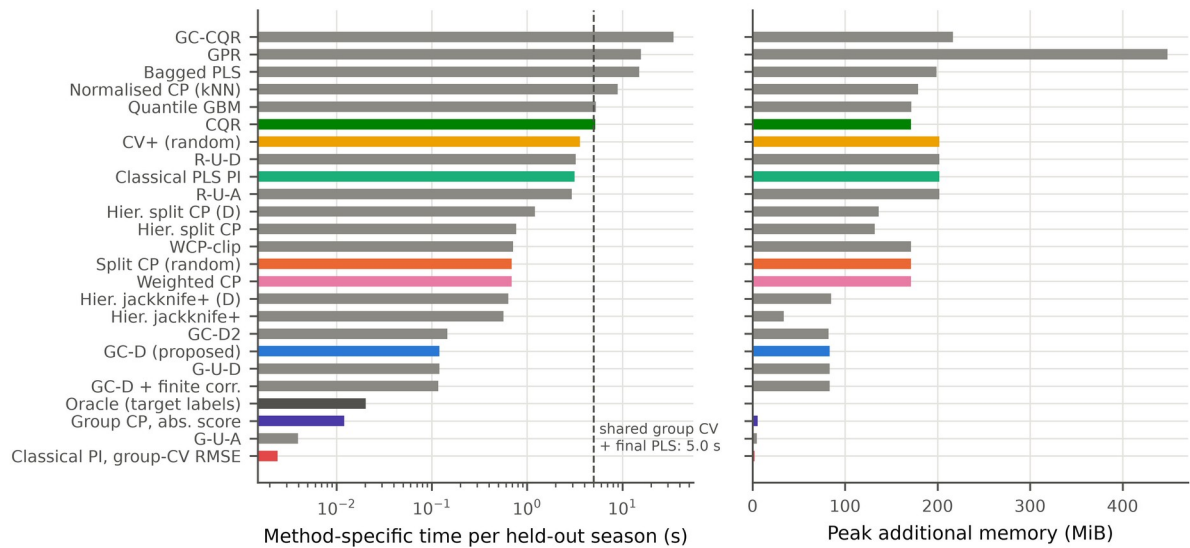


Fig. S2. Method-specific computing time (left, log scale) and peak additional memory (right); dashed line: shared group-wise cross-validation and final PLS fit.

S10. Pooled coverage with cluster-bootstrap confidence intervals

Table S10. Coverage pooled over all test spectra with 95 % cluster-bootstrap intervals (clusters = fruit or soil sample; seed 0).

task	method	pooled_cover age	ci_lo	ci_hi	n_spectra	n_units
mango_season_loso	ASTM-R	0.870	0.864	0.876	85401	10560
mango_season_loso	BAG	0.869	0.863	0.875	85401	10560
mango_season_loso	CQR	0.828	0.822	0.835	85401	10560
mango_season_loso	CV+	0.863	0.857	0.869	85401	10560
mango_season_loso	G-W-A	0.926	0.921	0.931	85401	10560
mango_season_loso	GC-D	0.927	0.923	0.932	85401	10560
mango_season_loso	GPR	0.853	0.847	0.860	85401	10560
mango_season_loso	ORACLE	0.899	0.894	0.904	85401	10560
mango_season_loso	SCP	0.859	0.853	0.865	85401	10560
mango_season_loso	WCP	0.878	0.872	0.884	85401	10560
mango_instrument	ASTM-R	0.836	0.830	0.842	82383	10508
mango_instrument	BAG	0.838	0.832	0.844	82383	10508
mango_instrument	CQR	0.796	0.789	0.803	82383	10508
mango_instrument	CV+	0.828	0.822	0.834	82383	10508
mango_instrument	G-W-A	0.906	0.902	0.911	82383	10508
mango_instrument	GC-D	0.911	0.906	0.915	82383	10508
mango_instrument	GPR	0.785	0.779	0.791	82383	10508
mango_instrument	ORACLE	0.900	0.894	0.905	82383	10508
mango_instrument	SCP	0.827	0.821	0.834	82383	10508
mango_instrument	WCP	0.919	0.914	0.924	82383	10508
mango_population	ASTM-R	0.892	0.886	0.898	85401	10560
mango_population	BAG	0.892	0.886	0.898	85401	10560
mango_population	CQR	0.855	0.848	0.861	85401	10560
mango_population	CV+	0.886	0.880	0.892	85401	10560
mango_population	G-W-A	0.898	0.893	0.904	85401	10560
mango_population	GC-D	0.896	0.891	0.902	85401	10560
mango_population	GPR	0.875	0.869	0.882	85401	10560
mango_population	ORACLE	0.897	0.892	0.901	85401	10560
mango_population	SCP	0.882	0.876	0.888	85401	10560
mango_population	WCP	0.908	0.902	0.914	85401	10560
ossl_lucas_block	ASTM-R	0.894	0.891	0.897	40175	40175
ossl_lucas_block	BAG	0.894	0.891	0.897	40175	40175
ossl_lucas_block	CQR	0.886	0.883	0.889	40175	40175
ossl_lucas_block	CV+	0.888	0.885	0.891	40175	40175
ossl_lucas_block	G-W-A	0.902	0.900	0.905	40175	40175
ossl_lucas_block	GC-D	0.898	0.895	0.901	40175	40175
ossl_lucas_block	GPR	0.888	0.885	0.891	40175	40175
ossl_lucas_block	ORACLE	0.901	0.898	0.904	40175	40175
ossl_lucas_block	SCP	0.886	0.883	0.889	40175	40175
ossl_lucas_block	WCP	0.883	0.880	0.886	40175	40175
mango_season_forward	ASTM-R	0.828	0.821	0.835	54891	7711

task	method	pooled_cover age	ci_lo	ci_hi	n_spectra	n_units
mango_season_forward	BAG	0.832	0.825	0.838	54891	7711
mango_season_forward	CQR	0.769	0.760	0.779	54891	7711
mango_season_forward	CV+	0.820	0.813	0.827	54891	7711
mango_season_forward	G-W-A	0.903	0.897	0.908	54891	7711
mango_season_forward	GC-D	0.920	0.914	0.925	54891	7711
mango_season_forward	GPR	0.799	0.791	0.807	54891	7711
mango_season_forward	ORACLE	0.900	0.895	0.905	54891	7711
mango_season_forward	SCP	0.822	0.815	0.829	54891	7711
mango_season_forward	WCP	0.911	0.904	0.916	54891	7711

S11. Specification-limit decisions

Table S11. Decisions 'DM $\geq$ L' accepted when the lower interval limit $\geq$ L (mango, new season, seed 0).

limit	method	n	accepted	false_accept_ among_accep ted	false_accept_ overall	yield_of_com pliant
14.000	ASTM-R	85401	0.476	0.022	0.011	0.616
14.000	BAG	85401	0.478	0.022	0.011	0.618
14.000	CQR	85401	0.517	0.027	0.014	0.665
14.000	CV+	85401	0.484	0.023	0.011	0.625
14.000	G-W-A	85401	0.411	0.013	0.006	0.536
14.000	GC-D	85401	0.422	0.013	0.005	0.551
14.000	GPR	85401	0.492	0.023	0.011	0.636
14.000	ORACLE	85401	0.454	0.015	0.007	0.591
14.000	SCP	85401	0.488	0.023	0.011	0.631
14.000	WCP	85401	0.475	0.020	0.009	0.615
15.000	ASTM-R	85401	0.324	0.022	0.007	0.520
15.000	BAG	85401	0.327	0.022	0.007	0.524
15.000	CQR	85401	0.331	0.013	0.004	0.535
15.000	CV+	85401	0.331	0.023	0.008	0.529
15.000	G-W-A	85401	0.271	0.015	0.004	0.438
15.000	GC-D	85401	0.279	0.009	0.003	0.453
15.000	GPR	85401	0.339	0.015	0.005	0.547
15.000	ORACLE	85401	0.301	0.016	0.005	0.485
15.000	SCP	85401	0.335	0.024	0.008	0.536
15.000	WCP	85401	0.319	0.017	0.006	0.513
16.000	ASTM-R	85401	0.204	0.039	0.008	0.416
16.000	BAG	85401	0.205	0.039	0.008	0.419
16.000	CQR	85401	0.168	0.017	0.003	0.351
16.000	CV+	85401	0.208	0.039	0.008	0.426
16.000	G-W-A	85401	0.161	0.030	0.005	0.332
16.000	GC-D	85401	0.164	0.021	0.003	0.342
16.000	GPR	85401	0.214	0.019	0.004	0.446
16.000	ORACLE	85401	0.181	0.026	0.005	0.375

limit	method	n	accepted	false_accept_ among_accep ted	false_accept_ overall	yield_of_com pliant
16.000	SCP	85401	0.213	0.041	0.009	0.434
16.000	WCP	85401	0.196	0.030	0.006	0.404
17.000	ASTM-R	85401	0.111	0.056	0.006	0.318
17.000	BAG	85401	0.112	0.055	0.006	0.322
17.000	CQR	85401	0.048	0.007	0.000	0.146
17.000	CV+	85401	0.114	0.056	0.006	0.329
17.000	G-W-A	85401	0.082	0.044	0.004	0.237
17.000	GC-D	85401	0.081	0.024	0.002	0.240
17.000	GPR	85401	0.115	0.034	0.004	0.338
17.000	ORACLE	85401	0.093	0.035	0.003	0.272
17.000	SCP	85401	0.118	0.057	0.007	0.339
17.000	WCP	85401	0.105	0.042	0.004	0.307

S12. Informativeness of the corn models on a new instrument

Table S12. RMSEP and bias on the held-out instrument vs. the SD of the reference values, and CQR width vs. the central 90 % range of the reference values (seed 0).

property	test_instrument	rmsep	bias	sd_reference	cqr_width	marginal_90 _range
moisture	m5	0.931	0.893	0.380	1.232	1.185
moisture	mp5	0.345	-0.148	0.380	1.240	1.185
moisture	mp6	0.313	0.018	0.380	1.399	1.185
oil	m5	0.153	-0.071	0.177	0.534	0.567
oil	mp5	0.173	0.047	0.177	0.590	0.567
oil	mp6	0.172	0.067	0.177	0.572	0.567
protein	m5	0.456	0.408	0.499	1.215	1.583
protein	mp5	0.398	0.074	0.499	1.486	1.583
protein	mp6	0.304	-0.126	0.499	1.406	1.583
starch	m5	1.315	-1.263	0.821	2.121	2.557
starch	mp5	0.829	0.121	0.821	2.621	2.557
starch	mp6	0.683	0.186	0.821	2.359	2.557

S13. Number of calibration groups

Table S13. Coverage when only K training groups are used (three repetitions); finite = minimum share of bounded intervals.

task	method	K	coverage	coverage_sd	width	finite
mango_instrument	CV+	3	0.636	0.085	4.016	1.000
mango_instrument	CV+	5	0.675	0.031	4.238	1.000
mango_instrument	CV+	10	0.727	0.052	3.966	1.000
mango_instrument	CV+	20	0.755	0.019	3.791	1.000
mango_instrument	G-W-A	3	0.786	0.078	5.579	1.000
mango_instrument	G-W-A	5	0.800	0.007	5.520	1.000

task	method	K	coverage	coverage_sd	width	finite
mango_instrument	G-W-A	10	0.868	0.045	5.433	1.000
mango_instrument	G-W-A	20	0.866	0.013	4.994	1.000
mango_instrument	GC-D	3	0.757	0.068	5.474	1.000
mango_instrument	GC-D	5	0.782	0.019	5.627	1.000
mango_instrument	GC-D	10	0.861	0.030	5.327	1.000
mango_instrument	GC-D	20	0.865	0.011	4.854	1.000
mango_instrument	HCP-D	3	1.000	0.000	∞	0.000
mango_instrument	HCP-D	5	1.000	0.000	∞	0.000
mango_instrument	HCP-D	10	1.000	0.000	∞	0.000
mango_instrument	HCP-D	20	0.972	0.022	17.622	1.000
mango_instrument	HJ+	3	1.000	0.000	∞	0.000
mango_instrument	HJ+	5	1.000	0.000	∞	0.000
mango_instrument	HJ+	10	0.987	0.005	10.153	1.000
mango_instrument	HJ+	20	0.942	0.003	6.142	1.000
mango_instrument	ORACLE	3	0.899	0.001	7.512	1.000
mango_instrument	ORACLE	5	0.900	0.001	8.310	1.000
mango_instrument	ORACLE	10	0.900	0.000	5.405	1.000
mango_instrument	ORACLE	20	0.900	0.001	5.062	1.000
mango_instrument	SCP	3	0.627	0.085	3.981	1.000
mango_instrument	SCP	5	0.672	0.036	4.227	1.000
mango_instrument	SCP	10	0.725	0.049	3.968	1.000
mango_instrument	SCP	20	0.754	0.022	3.779	1.000
ossl_lucas_block	CV+	3	0.805	0.013	1182.591	1.000
ossl_lucas_block	CV+	5	0.834	0.011	75.373	1.000
ossl_lucas_block	CV+	10	0.868	0.013	366.406	1.000
ossl_lucas_block	CV+	20	0.877	0.002	6.112	1.000
ossl_lucas_block	G-W-A	3	0.877	0.016	452.973	1.000
ossl_lucas_block	G-W-A	5	0.887	0.009	18.926	1.000
ossl_lucas_block	G-W-A	10	0.900	0.010	15.037	1.000
ossl_lucas_block	G-W-A	20	0.898	0.003	6.638	1.000
ossl_lucas_block	GC-D	3	0.864	0.012	7681.947	1.000
ossl_lucas_block	GC-D	5	0.876	0.007	60.569	1.000
ossl_lucas_block	GC-D	10	0.902	0.008	45.105	1.000
ossl_lucas_block	GC-D	20	0.898	0.002	7.453	1.000
ossl_lucas_block	HCP-D	3	1.000	0.000	∞	0.000
ossl_lucas_block	HCP-D	5	1.000	0.000	∞	0.000
ossl_lucas_block	HCP-D	10	1.000	0.000	∞	0.000
ossl_lucas_block	HCP-D	20	0.990	0.002	12281024863 2.972	1.000
ossl_lucas_block	HJ+	3	1.000	0.000	∞	0.000
ossl_lucas_block	HJ+	5	1.000	0.000	∞	0.000
ossl_lucas_block	HJ+	10	0.989	0.001	8892.885	1.000
ossl_lucas_block	HJ+	20	0.945	0.001	8.502	1.000
ossl_lucas_block	ORACLE	3	0.902	0.001	529.612	1.000
ossl_lucas_block	ORACLE	5	0.901	0.001	20.594	1.000

task	method	K	coverage	coverage_sd	width	finite
ossl_lucas_block	ORACLE	10	0.902	0.001	17.834	1.000
ossl_lucas_block	ORACLE	20	0.901	0.000	7.274	1.000
ossl_lucas_block	SCP	3	0.803	0.010	14.950	1.000
ossl_lucas_block	SCP	5	0.833	0.012	13.326	1.000
ossl_lucas_block	SCP	10	0.869	0.013	15.605	1.000
ossl_lucas_block	SCP	20	0.876	0.002	6.081	1.000

S14. Paired comparisons

Table S14. All paired comparisons of GC-D with each comparator (two-sided Wilcoxon; Holm within scenario and, more conservatively, across all main-scenario tests).

task	comparator	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
corn_moisture	ASTM-G	held-out group	30	0.0748	0.1398	0.5038	1.0000	1.0000	0.0000	0.2422
corn_moisture	ASTM-R	held-out group	30	0.0859	0.0839	0.6884	1.0000	1.0000	0.0000	0.2422
corn_moisture	BAG	held-out group	30	0.1250	0.3548	0.0897	0.9869	1.0000	0.0150	0.1609
corn_moisture	CQR	held-out group	30	0.8063	0.7677	0.0002	0.0031	0.0340	0.2750	0.0002
corn_moisture	CV+	held-out group	30	0.0882	0.1699	0.4165	1.0000	1.0000	0.0100	0.2321
corn_moisture	G-U-A	held-out group	30	0.0403	0.0280	0.8936	1.0000	1.0000	0.0000	0.2422
corn_moisture	G-U-D	held-out group	30	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000	1.0000
corn_moisture	G-W-A	held-out group	30	0.0403	0.0280	0.8936	1.0000	1.0000	0.0000	0.2422
corn_moisture	GC-D+	held-out group	30	- 999999997 .5609	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075
corn_moisture	GPR	held-out group	30	0.5050	0.7720	0.0002	0.0031	0.0316	0.0850	0.0021
corn_moisture	HCP-A	held-out group	30	- 999999997 .5609	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075
corn_moisture	HCP-D	held-out group	30	- 999999997 .5609	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075
corn_moisture	HJ+	held-out group	30	- 999999997 .5609	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075
corn_moisture	HJ+-D	held-out group	30	- 999999997 .5609	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075
corn_moisture	NCP-kNN	held-out group	30	-0.0029	0.0452	0.8290	1.0000	1.0000	0.0200	0.1145

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
corn_moisture	QGB	held-out group	30	0.5840	0.6946	0.0009	0.0107	0.1234	0.1300	0.0248
corn_moisture	R-U-A	held-out group	30	0.0381	-0.0237	0.9099	1.0000	1.0000	0.0050	0.2321
corn_moisture	R-U-D	held-out group	30	0.1197	0.1570	0.4528	1.0000	1.0000	0.0000	0.2321
corn_moisture	SCP	held-out group	30	0.0727	0.3247	0.1204	1.0000	1.0000	0.0200	0.0139
corn_moisture	WCP	held-out group	30	- 999999997 .3841	-0.9871	0.0000	0.0000	0.0004	0.3000	0.0040
corn_oil	ASTM-G	held-out group	30	0.0272	0.3075	0.1414	1.0000	1.0000	0.0600	0.2347
corn_oil	ASTM-R	held-out group	30	0.0053	0.0968	0.6435	1.0000	1.0000	0.0050	1.0000
corn_oil	BAG	held-out group	30	0.0242	0.4194	0.0449	0.5390	1.0000	0.0450	0.2761
corn_oil	CQR	held-out group	30	0.1022	0.6559	0.0017	0.0239	0.2273	0.0850	0.0088
corn_oil	CV+	held-out group	30	0.0314	0.2903	0.1650	1.0000	1.0000	0.0250	1.0000
corn_oil	G-U-A	held-out group	30	0.0009	0.0925	0.6583	1.0000	1.0000	0.0550	0.2700
corn_oil	G-U-D	held-out group	30	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000	1.0000
corn_oil	G-W-A	held-out group	30	0.0009	0.0925	0.6583	1.0000	1.0000	0.0550	0.2700
corn_oil	GC-D+	held-out group	30	- 999999999 .2904	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	GPR	held-out group	30	0.0535	0.4022	0.0545	0.5991	1.0000	0.0600	0.3529
corn_oil	HCP-A	held-out group	30	- 999999999 .2904	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	HCP-D	held-out group	30	- 999999999 .2904	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	HJ+	held-out group	30	- 999999999 .2904	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	HJ+-D	held-out group	30	- 999999999 .2904	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	NCP-kNN	held-out group	30	-0.0474	-0.3247	0.1204	1.0000	1.0000	0.0100	1.0000
corn_oil	QGB	held-out group	30	0.0113	0.0882	0.6733	1.0000	1.0000	-0.0550	1.0000
corn_oil	R-U-A	held-out	30	-0.0085	0.0237	0.9099	1.0000	1.0000	0.0000	1.0000

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
corn_oil	R-U-D	group held-out group	30	0.0639	0.4925	0.0185	0.2407	1.0000	0.0150	0.4197
corn_oil	SCP	group held-out group	30	-0.0359	-0.2258	0.2802	1.0000	1.0000	0.0350	1.0000
corn_oil	WCP	group held-out group	30	- 999999999 .2429	-0.9871	0.0000	0.0000	0.0004	0.0450	0.0604
corn_protein	ASTM-G	group held-out group	30	0.1044	0.2817	0.1779	1.0000	1.0000	-0.0100	1.0000
corn_protein	ASTM-R	group held-out group	30	0.1780	0.1484	0.4779	1.0000	1.0000	-0.0050	1.0000
corn_protein	BAG	group held-out group	30	0.1267	0.2086	0.3185	1.0000	1.0000	0.0050	1.0000
corn_protein	CQR	group held-out group	30	0.3458	0.6946	0.0009	0.0125	0.1234	0.1250	0.0014
corn_protein	CV+	group held-out group	30	0.1527	0.1054	0.6143	1.0000	1.0000	-0.0050	1.0000
corn_protein	G-U-A	group held-out group	30	0.0920	0.0753	0.7189	1.0000	1.0000	-0.0050	1.0000
corn_protein	G-U-D	group held-out group	30	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000	1.0000
corn_protein	G-W-A	group held-out group	30	0.0920	0.0753	0.7189	1.0000	1.0000	-0.0050	1.0000
corn_protein	GC-D+	group held-out group	30	- 999999998 .1847	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	GPR	group held-out group	30	0.2833	0.6645	0.0015	0.0193	0.1988	0.0250	1.0000
corn_protein	HCP-A	group held-out group	30	- 999999998 .1847	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	HCP-D	group held-out group	30	- 999999998 .1847	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	HJ+	group held-out group	30	- 999999998 .1847	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	HJ+-D	group held-out group	30	- 999999998 .1847	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	NCP-kNN	group held-out group	30	0.0470	-0.0968	0.6435	1.0000	1.0000	0.0000	1.0000
corn_protein	QGB	group held-out group	30	0.0569	0.2258	0.2802	1.0000	1.0000	0.0000	1.0000
corn_protein	R-U-A	group held-out group	30	0.1778	0.0323	0.8774	1.0000	1.0000	-0.0100	1.0000
corn_protein	R-U-D	group held-out group	30	0.1248	0.4237	0.0428	0.5132	1.0000	0.0150	1.0000

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
corn_protein	SCP	held-out group	30	0.0454	-0.0968	0.6435	1.0000	1.0000	0.0000	1.0000
corn_protein	WCP	held-out group	30	- 999999998 .1847	-1.0000	0.0000	0.0000	0.0003	0.0850	0.0103
corn_starch	ASTM-G	held-out group	30	0.4593	0.3806	0.0687	0.7221	1.0000	0.0050	1.0000
corn_starch	ASTM-R	held-out group	30	0.3343	0.1484	0.4779	1.0000	1.0000	0.0000	1.0000
corn_starch	BAG	held-out group	30	0.7404	0.4581	0.0285	0.3418	1.0000	0.0150	1.0000
corn_starch	CQR	held-out group	30	0.7189	0.8151	0.0001	0.0014	0.0142	0.0500	0.0064
corn_starch	CV+	held-out group	30	0.3772	0.1484	0.4779	1.0000	1.0000	0.0100	1.0000
corn_starch	G-U-A	held-out group	30	0.3859	0.1957	0.3493	1.0000	1.0000	0.0000	1.0000
corn_starch	G-U-D	held-out group	30	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000	1.0000
corn_starch	G-W-A	held-out group	30	0.3859	0.1957	0.3493	1.0000	1.0000	0.0000	1.0000
corn_starch	GC-D+	held-out group	30	- 999999995 .6617	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	GPR	held-out group	30	0.4886	0.2602	0.2134	1.0000	1.0000	-0.0050	1.0000
corn_starch	HCP-A	held-out group	30	- 999999995 .6617	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	HCP-D	held-out group	30	- 999999995 .6617	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	HJ+	held-out group	30	- 999999995 .6617	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	HJ+-D	held-out group	30	- 999999995 .6617	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	NCP-kNN	held-out group	30	0.0716	-0.1656	0.4284	1.0000	1.0000	0.0000	1.0000
corn_starch	QGB	held-out group	30	0.2059	0.3849	0.0656	0.7221	1.0000	-0.0550	1.0000
corn_starch	R-U-A	held-out group	30	0.3156	0.0065	0.9754	1.0000	1.0000	0.0050	1.0000
corn_starch	R-U-D	held-out group	30	0.6903	0.6258	0.0028	0.0359	0.3567	0.0150	1.0000
corn_starch	SCP	held-out group	30	-0.0360	-0.2000	0.3389	1.0000	1.0000	-0.0250	1.0000
corn_starch	WCP	held-out	30	-	-0.9957	0.0000	0.0000	0.0003	0.0300	0.0432

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
		group		999999995 .6617						
mango_instrument	ASTM-G	held-out group	24	-0.1169	-0.3667	0.1208	1.0000	1.0000	-0.0014	1.0000
mango_instrument	ASTM-R	held-out group	24	-0.0729	-0.3067	0.1974	1.0000	1.0000	-0.0124	0.7900
mango_instrument	BAG	held-out group	24	-0.0629	-0.2867	0.2292	1.0000	1.0000	-0.0117	0.7900
mango_instrument	CQR	held-out group	24	-0.8632	-0.6467	0.0043	0.0650	0.5506	-0.0416	0.0866
mango_instrument	CV+	held-out group	24	-0.1091	-0.3867	0.1011	1.0000	1.0000	-0.0175	0.6798
mango_instrument	G-U-A	held-out group	24	-0.0252	-0.2067	0.3902	1.0000	1.0000	0.0029	1.0000
mango_instrument	G-U-D	held-out group	24	-0.0321	0.0000	1.0000	1.0000	1.0000	-0.0014	1.0000
mango_instrument	G-W-A	held-out group	24	-0.0537	-0.2733	0.2522	1.0000	1.0000	0.0021	1.0000
mango_instrument	GC-D+	held-out group	24	-0.1321	-0.2667	0.2643	1.0000	1.0000	-0.0103	1.0000
mango_instrument	GPR	held-out group	24	-0.5475	-0.5467	0.0179	0.2502	1.0000	-0.0500	0.1482
mango_instrument	HCP-A	held-out group	24	-6.7965	-1.0000	0.0000	0.0000	0.0000	-0.0296	1.0000
mango_instrument	HCP-D	held-out group	24	-4.8531	-1.0000	0.0000	0.0000	0.0000	-0.0177	1.0000
mango_instrument	HJ+	held-out group	24	-0.2407	-0.2800	0.2405	1.0000	1.0000	-0.0119	1.0000
mango_instrument	HJ+-D	held-out group	24	-0.1533	-0.3133	0.1875	1.0000	1.0000	-0.0126	1.0000
mango_instrument	NCP-kNN	held-out group	24	-0.6518	-0.8800	0.0000	0.0005	0.0045	-0.0571	0.0427
mango_instrument	QGB	held-out group	24	-1.3881	-0.7333	0.0010	0.0154	0.1309	-0.0889	0.0000
mango_instrument	R-U-A	held-out group	24	-0.1189	-0.3933	0.0951	1.0000	1.0000	-0.0177	0.6798
mango_instrument	R-U-D	held-out group	24	-0.0164	-0.1400	0.5646	1.0000	1.0000	-0.0160	1.0000
mango_instrument	SCP	held-out group	24	-0.1264	-0.3867	0.1011	1.0000	1.0000	-0.0167	0.7900
mango_instrument	WCP	held-out group	24	- 999999994 .4245	-1.0000	0.0000	0.0000	0.0000	0.0064	1.0000
mango_population	ASTM-G	outer fold (seed-0 partition)	10	-0.0471	-0.5636	0.1309	1.0000	1.0000	-0.0022	1.0000
mango_population	ASTM-R	outer fold (seed-0 partition)	10	-0.0477	-0.5636	0.1309	1.0000	1.0000	-0.0052	1.0000

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
mango_population	BAG	outer fold (seed-0 partition)	10	-0.0471	-0.5636	0.1309	1.0000	1.0000	-0.0044	1.0000
mango_population	CQR	outer fold (seed-0 partition)	10	-0.3762	-0.7091	0.0488	0.7324	1.0000	-0.0450	0.2598
mango_population	CV+	outer fold (seed-0 partition)	10	-0.0464	-0.6727	0.0645	0.7734	1.0000	-0.0069	1.0000
mango_population	G-U-A	outer fold (seed-0 partition)	10	-0.0454	-0.5636	0.1309	1.0000	1.0000	-0.0028	1.0000
mango_population	G-U-D	outer fold (seed-0 partition)	10	-0.0011	-0.0545	0.9219	1.0000	1.0000	-0.0010	1.0000
mango_population	G-W-A	outer fold (seed-0 partition)	10	-0.0449	-0.6364	0.0840	0.9238	1.0000	-0.0037	1.0000
mango_population	GC-D+	outer fold (seed-0 partition)	10	-0.0004	-0.1273	0.7695	1.0000	1.0000	-0.0009	1.0000
mango_population	GPR	outer fold (seed-0 partition)	10	0.2715	0.7455	0.0371	0.5938	1.0000	-0.0089	1.0000
mango_population	HCP-A	outer fold (seed-0 partition)	10	-0.1884	-0.8545	0.0137	0.2324	1.0000	-0.0035	1.0000
mango_population	HCP-D	outer fold (seed-0 partition)	10	-0.1645	-0.8909	0.0098	0.1758	1.0000	-0.0060	1.0000
mango_population	HJ+	outer fold (seed-0 partition)	10	-0.0486	-0.4182	0.2754	1.0000	1.0000	-0.0040	1.0000
mango_population	HJ+-D	outer fold (seed-0 partition)	10	-0.0056	0.1636	0.6953	1.0000	1.0000	-0.0020	1.0000
mango_population	NCP-kNN	outer fold (seed-0 partition)	10	0.2138	0.4182	0.2754	1.0000	1.0000	-0.0457	0.3516
mango_population	QGB	outer fold (seed-0 partition)	10	-0.7198	-0.9636	0.0039	0.0742	0.5000	-0.0960	0.0391
mango_population	R-U-A	outer fold (seed-0 partition)	10	-0.0467	-0.7091	0.0488	0.7324	1.0000	-0.0074	1.0000
mango_population	R-U-D	outer fold (seed-0 partition)	10	0.0018	0.1273	0.7695	1.0000	1.0000	-0.0041	1.0000
mango_population	SCP	outer fold (seed-0	10	-0.0527	-0.7091	0.0488	0.7324	1.0000	-0.0144	0.8301

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
mango_population	WCP	partition) outer fold (seed-0 partition)	10	- 999999995 .1468	-1.0000	0.0020	0.0391	0.2559	-0.0032	1.0000
mango_season_forwar d	ASTM-G	held-out group	5	-0.2464	-0.4667	0.4375	1.0000	1.0000	0.0011	1.0000
mango_season_forwar d	ASTM-R	held-out group	5	-0.0948	-0.4667	0.4375	1.0000	1.0000	-0.0725	1.0000
mango_season_forwar d	BAG	held-out group	5	-0.0583	-0.4667	0.4375	1.0000	1.0000	-0.0686	1.0000
mango_season_forwar d	CQR	held-out group	5	-1.1948	-1.0000	0.0625	1.0000	1.0000	-0.1282	1.0000
mango_season_forwar d	CV+	held-out group	5	-0.1315	-0.6000	0.3125	1.0000	1.0000	-0.0795	1.0000
mango_season_forwar d	G-U-A	held-out group	5	-0.1772	-0.4667	0.4375	1.0000	1.0000	-0.0059	1.0000
mango_season_forwar d	G-U-D	held-out group	5	-0.0383	-0.4667	0.4375	1.0000	1.0000	-0.0055	1.0000
mango_season_forwar d	G-W-A	held-out group	5	-0.1775	-0.4667	0.4375	1.0000	1.0000	0.0004	1.0000
mango_season_forwar d	GC-D+	held-out group	5	- 999999993 .6803	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forwar d	GPR	held-out group	5	-0.6755	-1.0000	0.0625	1.0000	1.0000	-0.1180	1.0000
mango_season_forwar d	HCP-A	held-out group	5	- 999999993 .6803	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forwar d	HCP-D	held-out group	5	- 999999993 .6803	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forwar d	HJ+	held-out group	5	- 999999993 .6803	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forwar d	HJ+-D	held-out group	5	- 999999993 .6803	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forwar d	NCP-kNN	held-out group	5	-0.6276	-0.6000	0.3125	1.0000	1.0000	-0.0675	1.0000
mango_season_forwar d	QGB	held-out group	5	-1.9253	-1.0000	0.0625	1.0000	1.0000	-0.1819	1.0000
mango_season_forwar d	R-U-A	held-out group	5	-0.1418	-0.6000	0.3125	1.0000	1.0000	-0.0828	1.0000
mango_season_forwar d	R-U-D	held-out group	5	-0.0353	-0.4667	0.4375	1.0000	1.0000	-0.0706	1.0000
mango_season_forwar d	SCP	held-out group	5	-0.0972	-0.4667	0.4375	1.0000	1.0000	-0.0767	1.0000
mango_season_forwar d	WCP	held-out group	5	-4.3649	-0.8667	0.1250	1.0000	1.0000	-0.0076	1.0000

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
mango_season_loso	ASTM-G	held-out group	7	-0.1824	-0.3571	0.4688	1.0000	1.0000	-0.0002	1.0000
mango_season_loso	ASTM-R	held-out group	7	-0.0494	-0.3571	0.4688	1.0000	1.0000	-0.0648	1.0000
mango_season_loso	BAG	held-out group	7	-0.0535	-0.3571	0.4688	1.0000	1.0000	-0.0648	1.0000
mango_season_loso	CQR	held-out group	7	-0.5826	-0.6429	0.1562	1.0000	1.0000	-0.1076	1.0000
mango_season_loso	CV+	held-out group	7	-0.0810	-0.4286	0.3750	1.0000	1.0000	-0.0744	1.0000
mango_season_loso	G-U-A	held-out group	7	0.0544	0.0000	1.0000	1.0000	1.0000	-0.0155	1.0000
mango_season_loso	G-U-D	held-out group	7	-0.0465	-0.2143	0.6875	1.0000	1.0000	-0.0184	1.0000
mango_season_loso	G-W-A	held-out group	7	-0.1016	-0.3571	0.4688	1.0000	1.0000	0.0015	1.0000
mango_season_loso	GC-D+	held-out group	7	- 999999993 .9316	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	GPR	held-out group	7	-0.6049	-0.5714	0.2188	1.0000	1.0000	-0.0791	1.0000
mango_season_loso	HCP-A	held-out group	7	- 999999993 .9316	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	HCP-D	held-out group	7	- 999999993 .9316	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	HJ+	held-out group	7	- 999999993 .9316	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	HJ+-D	held-out group	7	- 999999993 .9316	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	NCP-kNN	held-out group	7	-0.2430	-0.4286	0.3750	1.0000	1.0000	-0.0706	1.0000
mango_season_loso	QGB	held-out group	7	-1.0478	-0.8571	0.0469	0.7031	1.0000	-0.1675	0.6250
mango_season_loso	R-U-A	held-out group	7	-0.0812	-0.5000	0.2969	1.0000	1.0000	-0.0746	1.0000
mango_season_loso	R-U-D	held-out group	7	-0.0495	-0.2857	0.5781	1.0000	1.0000	-0.0479	1.0000
mango_season_loso	SCP	held-out group	7	-0.1138	-0.3571	0.4688	1.0000	1.0000	-0.0772	1.0000
mango_season_loso	WCP	held-out group	7	-0.4367	-0.5000	0.2969	1.0000	1.0000	-0.0110	1.0000
ossl_kssl_to_lucas	ASTM-G	outer fold	1	-30.7141	-1.0000	1.0000	1.0000	–	-0.5333	1.0000
ossl_kssl_to_lucas	ASTM-R	outer fold	1	-41.0577	-1.0000	1.0000	1.0000	–	-0.5512	1.0000
ossl_kssl_to_lucas	BAG	outer fold	1	-41.4047	-1.0000	1.0000	1.0000	–	-0.5493	1.0000

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
ossl_kssl_to_lucas	CQR	outer fold	1	-10.0420	-1.0000	1.0000	1.0000	–	-0.2944	1.0000
ossl_kssl_to_lucas	CV+	outer fold	1	-43.7012	-1.0000	1.0000	1.0000	–	-0.5542	1.0000
ossl_kssl_to_lucas	G-U-A	outer fold	1	-40.1325	-1.0000	1.0000	1.0000	–	-0.5498	1.0000
ossl_kssl_to_lucas	G-U-D	outer fold	1	2.2407	1.0000	1.0000	1.0000	–	-0.0537	1.0000
ossl_kssl_to_lucas	G-W-A	outer fold	1	-33.3959	-1.0000	1.0000	1.0000	–	-0.5384	1.0000
ossl_kssl_to_lucas	GC-D+	outer fold	1	-3.6857	-1.0000	1.0000	1.0000	–	0.0585	1.0000
ossl_kssl_to_lucas	GPR	outer fold	1	12.3398	1.0000	1.0000	1.0000	–	0.2205	1.0000
ossl_kssl_to_lucas	HCP-A	outer fold	1	-22.5906	-1.0000	1.0000	1.0000	–	-0.4366	1.0000
ossl_kssl_to_lucas	HCP-D	outer fold	1	-5.9800	-1.0000	1.0000	1.0000	–	0.1078	1.0000
ossl_kssl_to_lucas	HJ+	outer fold	1	-27.1396	-1.0000	1.0000	1.0000	–	-0.5262	1.0000
ossl_kssl_to_lucas	HJ+-D	outer fold	1	-3.5396	-1.0000	1.0000	1.0000	–	0.0652	1.0000
ossl_kssl_to_lucas	NCP-kNN	outer fold	1	-28.6438	-1.0000	1.0000	1.0000	–	-0.4900	1.0000
ossl_kssl_to_lucas	QGB	outer fold	1	-13.5568	-1.0000	1.0000	1.0000	–	-0.3150	1.0000
ossl_kssl_to_lucas	R-U-A	outer fold	1	-43.7225	-1.0000	1.0000	1.0000	–	-0.5544	1.0000
ossl_kssl_to_lucas	R-U-D	outer fold	1	3.8305	1.0000	1.0000	1.0000	–	-0.1103	1.0000
ossl_kssl_to_lucas	SCP	outer fold	1	-52.8624	-1.0000	1.0000	1.0000	–	-0.5580	1.0000
ossl_kssl_to_lucas	WCP	outer fold	1	-	-1.0000	1.0000	1.0000	–	0.1955	1.0000
				999999904 .4804						
ossl_lucas_block	ASTM-G	held-out group	30	-0.0701	-0.4624	0.0270	0.1622	1.0000	0.0026	1.0000
ossl_lucas_block	ASTM-R	held-out group	30	-0.1610	-0.7333	0.0005	0.0045	0.0630	0.0060	1.0000
ossl_lucas_block	BAG	held-out group	30	-0.1457	-0.6516	0.0018	0.0147	0.2419	0.0047	1.0000
ossl_lucas_block	CQR	held-out group	30	1.5369	1.0000	0.0000	0.0000	0.0003	0.0406	0.0412
ossl_lucas_block	CV+	held-out group	30	-0.1841	-0.7935	0.0001	0.0021	0.0214	0.0047	1.0000
ossl_lucas_block	G-U-A	held-out group	30	-0.1556	-0.6774	0.0012	0.0108	0.1616	0.0055	1.0000
ossl_lucas_block	G-U-D	held-out group	30	-0.0000	0.2215	0.2895	0.9477	1.0000	0.0000	1.0000
ossl_lucas_block	G-W-A	held-out group	30	-0.1585	-0.6387	0.0023	0.0158	0.2932	0.0037	1.0000
ossl_lucas_block	GC-D+	held-out group	30	-0.1610	-0.0710	0.7343	1.0000	1.0000	-0.0128	1.0000
ossl_lucas_block	GPR	held-out group	30	0.7942	0.9398	0.0000	0.0001	0.0010	0.0124	0.6321
ossl_lucas_block	HCP-A	held-out group	30	-1.2277	-0.7505	0.0003	0.0036	0.0464	-0.0251	1.0000
ossl_lucas_block	HCP-D	held-out group	30	-1.1193	-0.7935	0.0001	0.0021	0.0214	-0.0291	1.0000
ossl_lucas_block	HJ+	held-out group	30	-0.1840	-0.2473	0.2369	0.9477	1.0000	-0.0087	1.0000
ossl_lucas_block	HJ+-D	held-out group	30	-0.1200	-0.1011	0.6288	1.0000	1.0000	-0.0134	1.0000

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
ossl_lucas_block	NCP-kNN	held-out group	30	0.9220	0.9828	0.0000	0.0000	0.0004	0.0342	0.7326
ossl_lucas_block	QGB	held-out group	30	1.1799	0.9957	0.0000	0.0000	0.0003	-0.0017	1.0000
ossl_lucas_block	R-U-A	held-out group	30	-0.1888	-0.7935	0.0001	0.0021	0.0214	0.0056	1.0000
ossl_lucas_block	R-U-D	held-out group	30	0.0041	-0.2860	0.1714	0.8569	1.0000	0.0041	1.0000
ossl_lucas_block	SCP	held-out group	30	-0.1912	-0.8280	0.0001	0.0011	0.0110	0.0063	1.0000
ossl_lucas_block	WCP	held-out group	30	-	-0.9140	0.0000	0.0002	0.0018	-0.0023	1.0000
				999999973						
				.8658						
ossl_lucas_campaign	ASTM-G	outer fold	2	-0.5270	-1.0000	0.5000	1.0000	–	-0.0091	1.0000
ossl_lucas_campaign	ASTM-R	outer fold	2	-0.8751	-1.0000	0.5000	1.0000	–	-0.0089	1.0000
ossl_lucas_campaign	BAG	outer fold	2	-0.8964	-1.0000	0.5000	1.0000	–	-0.0082	1.0000
ossl_lucas_campaign	CQR	outer fold	2	2.4328	1.0000	0.5000	1.0000	–	0.0028	1.0000
ossl_lucas_campaign	CV+	outer fold	2	-1.0897	-1.0000	0.5000	1.0000	–	-0.0170	1.0000
ossl_lucas_campaign	G-U-A	outer fold	2	-0.7462	-1.0000	0.5000	1.0000	–	-0.0046	1.0000
ossl_lucas_campaign	G-U-D	outer fold	2	0.0094	0.3333	1.0000	1.0000	–	0.0059	1.0000
ossl_lucas_campaign	G-W-A	outer fold	2	-0.6831	-1.0000	0.5000	1.0000	–	-0.0076	1.0000
ossl_lucas_campaign	GC-D+	outer fold	2	0.3831	1.0000	0.5000	1.0000	–	0.0041	1.0000
ossl_lucas_campaign	GPR	outer fold	2	-0.2450	-0.3333	1.0000	1.0000	–	-0.0655	1.0000
ossl_lucas_campaign	HCP-A	outer fold	2	0.0004	0.3333	1.0000	1.0000	–	-0.0136	1.0000
ossl_lucas_campaign	HCP-D	outer fold	2	-0.4856	-0.3333	1.0000	1.0000	–	-0.0260	1.0000
ossl_lucas_campaign	HJ+	outer fold	2	0.0808	1.0000	0.5000	1.0000	–	-0.0022	1.0000
ossl_lucas_campaign	HJ+-D	outer fold	2	0.3615	1.0000	0.5000	1.0000	–	0.0034	1.0000
ossl_lucas_campaign	NCP-kNN	outer fold	2	0.5545	1.0000	0.5000	1.0000	–	-0.0711	1.0000
ossl_lucas_campaign	QGB	outer fold	2	1.4254	0.3333	1.0000	1.0000	–	-0.0623	1.0000
ossl_lucas_campaign	R-U-A	outer fold	2	-1.0909	-1.0000	0.5000	1.0000	–	-0.0179	1.0000
ossl_lucas_campaign	R-U-D	outer fold	2	-0.0877	-1.0000	0.5000	1.0000	–	0.0064	1.0000
ossl_lucas_campaign	SCP	outer fold	2	-1.3354	-1.0000	0.5000	1.0000	–	-0.0227	1.0000
ossl_lucas_campaign	WCP	outer fold	2	-	-1.0000	0.5000	1.0000	–	-0.0689	1.0000
				999999987						
				.8030						
tablets	ASTM-G	held-out group	2	-1.1514	-1.0000	0.5000	1.0000	–	-0.0191	1.0000
tablets	ASTM-R	held-out group	2	-1.1514	-1.0000	0.5000	1.0000	–	-0.0191	1.0000
tablets	BAG	held-out group	2	-1.2932	-1.0000	0.5000	1.0000	–	-0.0196	1.0000
tablets	CQR	held-out group	2	-13.5524	-1.0000	0.5000	1.0000	–	-0.0176	1.0000
tablets	CV+	held-out group	2	-0.9594	-1.0000	0.5000	1.0000	–	-0.0352	1.0000
tablets	G-U-A	held-out	2	-1.3245	-0.3333	1.0000	1.0000	–	-0.0489	1.0000

task	comparat or	unit	n_units	IS_media n_diff	IS_r_rb	IS_p	IS_p_hol m	IS_p_hol m_all_mai n	covgap_m edian_diff	covgap_p_ holm
		group								
tablets	G-U-D	held-out group	2	0.0000	0.0000	1.0000	1.0000	–	0.0000	1.0000
tablets	G-W-A	held-out group	2	-1.3245	-0.3333	1.0000	1.0000	–	-0.0489	1.0000
tablets	GC-D+	held-out group	2	-0.0156	-0.3333	1.0000	1.0000	–	-0.0002	1.0000
tablets	GPR	held-out group	2	-3.4028	-1.0000	0.5000	1.0000	–	-0.0830	1.0000
tablets	HCP-A	held-out group	2	-4.9206	-1.0000	0.5000	1.0000	–	-0.0759	1.0000
tablets	HCP-D	held-out group	2	-3.8605	-1.0000	0.5000	1.0000	–	-0.0128	1.0000
tablets	HJ+	held-out group	2	-0.9594	-1.0000	0.5000	1.0000	–	-0.0352	1.0000
tablets	HJ+-D	held-out group	2	-0.0208	-0.3333	1.0000	1.0000	–	-0.0002	1.0000
tablets	NCP-kNN	held-out group	2	-1.1166	-1.0000	0.5000	1.0000	–	-0.0157	1.0000
tablets	QGB	held-out group	2	-13.8898	-1.0000	0.5000	1.0000	–	-0.0578	1.0000
tablets	R-U-A	held-out group	2	-1.3245	-0.3333	1.0000	1.0000	–	-0.0489	1.0000
tablets	R-U-D	held-out group	2	0.0000	0.0000	1.0000	1.0000	–	0.0000	1.0000
tablets	SCP	held-out group	2	-2.4731	-1.0000	0.5000	1.0000	–	-0.0630	1.0000
tablets	WCP	held-out group	2	- 4999999988 .8037	-1.0000	0.5000	1.0000	–	-0.0385	1.0000

S15. Leave-one-cultivar-out test

Table S15a. Leave-one-cultivar-out (mango; ten cultivars; seeds 0–2; nominal 0.90).

method	coverage	coverage_mi n	width	interval_scor e	n_cultivars	n_seeds
ASTM-G	0.907	0.805	3.593	4.699	10	3
ASTM-R	0.907	0.788	3.589	4.703	10	3
CQR	0.855	0.738	3.604	5.367	10	3
CV+	0.902	0.776	3.524	4.704	10	3
G-W-A	0.901	0.786	3.509	4.710	10	3
GC-D	0.902	0.778	3.541	4.679	10	3
HJ+	1.000	1.000	20.604	20.604	10	3
ORACLE	0.898	0.892	3.419	4.478	10	3
SCP	0.900	0.772	3.504	4.712	10	3

Table S15b. Paired comparisons of GC-D with each comparator over cultivars (two-sided Wilcoxon, Holm correction).

comparator	n	IS_median_diff	IS_r_rb	IS_p	covgap_median_diff	covgap_p	IS_p_holm	covgap_p_holm
ASTM-G	10	-0.0807	-0.1636	0.6953	0.0073	0.1934	1.0000	0.9668
ASTM-R	10	-0.0722	-0.1273	0.7695	0.0103	0.1055	1.0000	0.6328
CQR	10	-0.6484	-0.8182	0.0195	-0.0064	0.6250	0.1172	1.0000
CV+	10	-0.0796	-0.3091	0.4316	0.0071	0.2324	1.0000	0.9668
G-W-A	10	-0.0997	-0.1636	0.6953	0.0008	0.4922	1.0000	1.0000
HJ+	10	-16.5582	-1.0000	0.0020	-0.0430	0.0098	0.0137	0.0684
SCP	10	-0.0699	-0.3455	0.3750	0.0027	0.4316	1.0000	1.0000

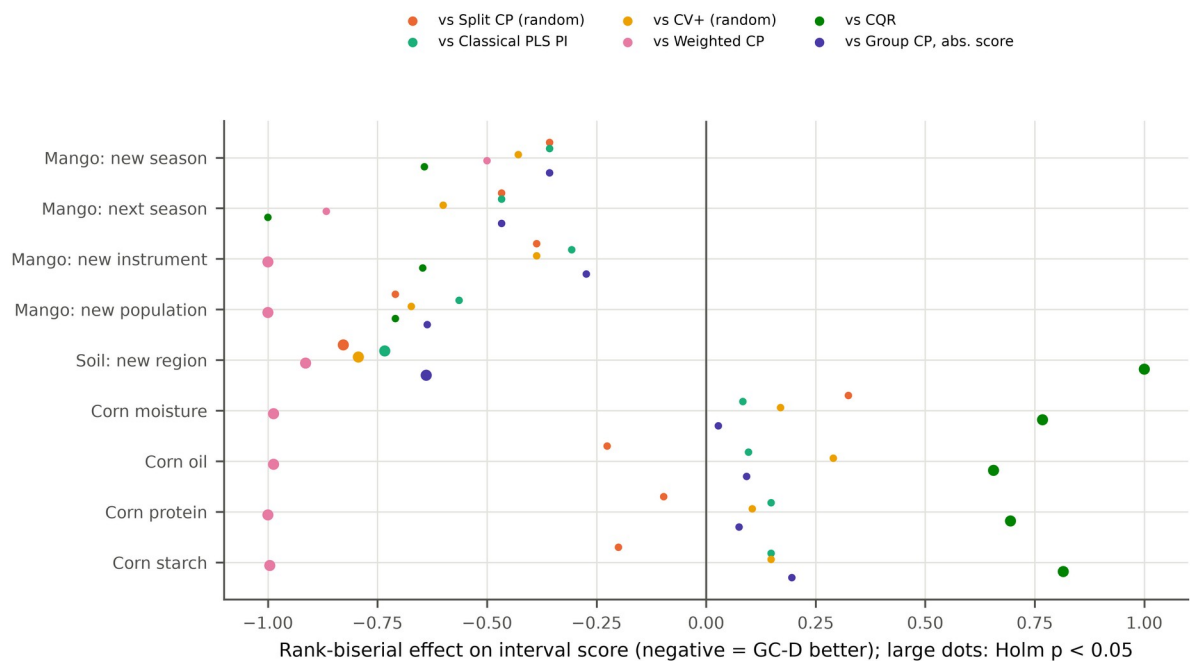


Fig. S1. Rank-biserial effect sizes of the interval-score differences between GC-D and selected comparators (negative: GC-D lower); larger symbols: Holm-adjusted $p < 0.05$.